Lesson 2: complex circuits – amplifiers – operational amplifier basics

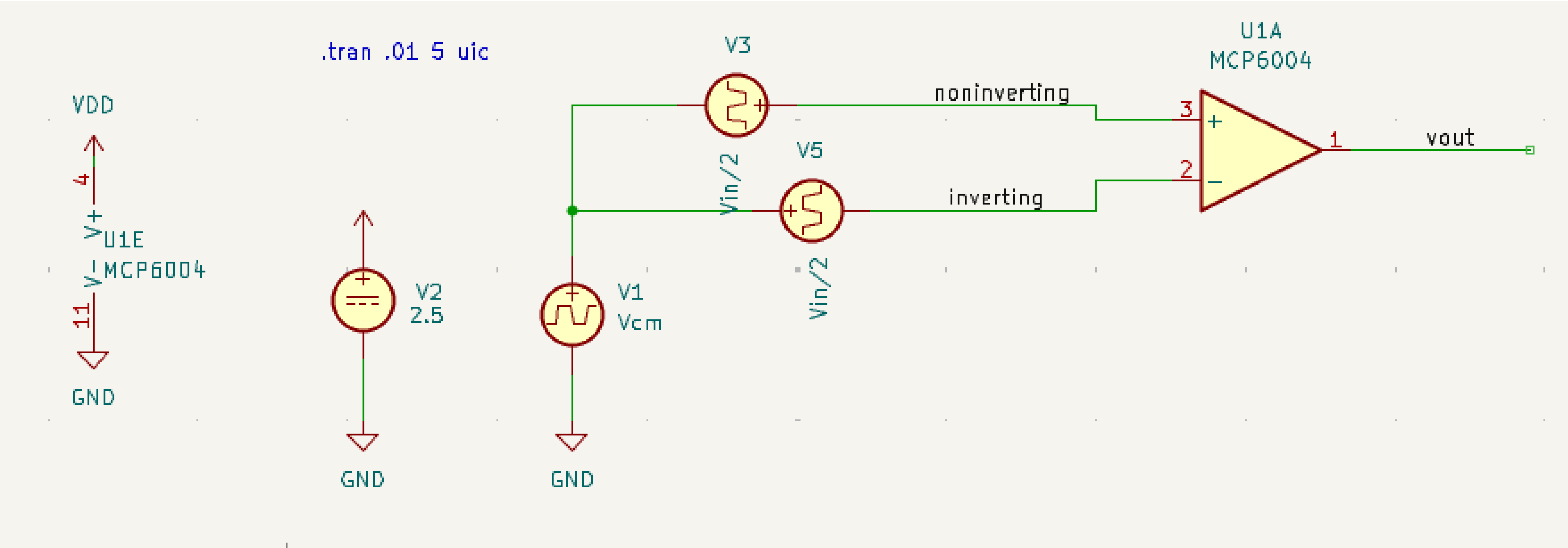
Neurons are more than linear integrators. Their voltage-gates ion channels create complex dynamics.

To replicate neural dynamics with analog electronic circuits, we need a way to change the voltage of our sources during operation, using *controlled sources* .

A voltage controlled voltage source is the operational amplifier, short OA. Our neuron will use the MC6004 integrated operational amplifier.

KC4: MC6004 is an operational amplifier (OA).

It hast an output V_{out} and two inputs $V_{noninverting}$ (short +) and $V_{inverting}$ (-).



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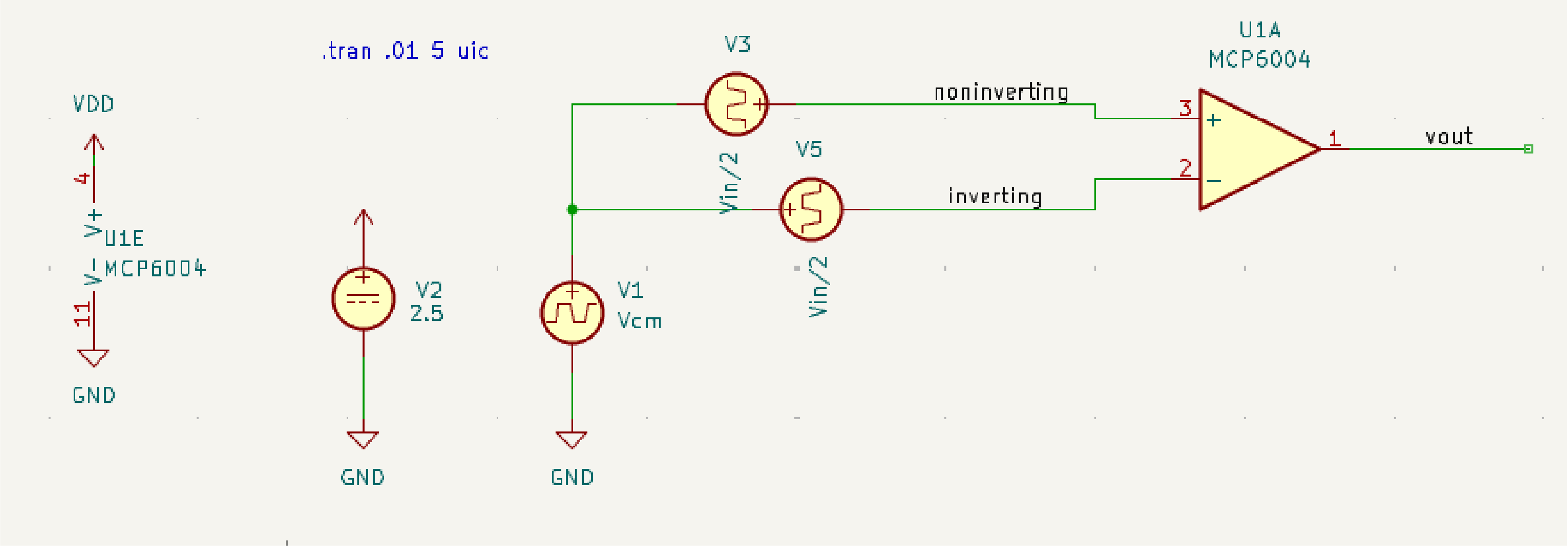


Q4.1: Verify that it amplifies $V_{noninverting} - V_{inverting}$, i.e. $V_{out}(V_{in}) = A * (V_{noninveriting} - V_{inverting})$.
Determine A from the simulation.

technical hints:

- you have to copy the `/shares/zitipoolhome/nca-opt/models` directory to your KiCad directory (where your .kicad_pro and .kicad_sch files are)
- you may use `/shares/zitipoolhome/nca-opt/2_opamp_basic.kicad_pro` and `/shares/zitipoolhome/nca-opt/2_opamp_basic.kicad_sch`

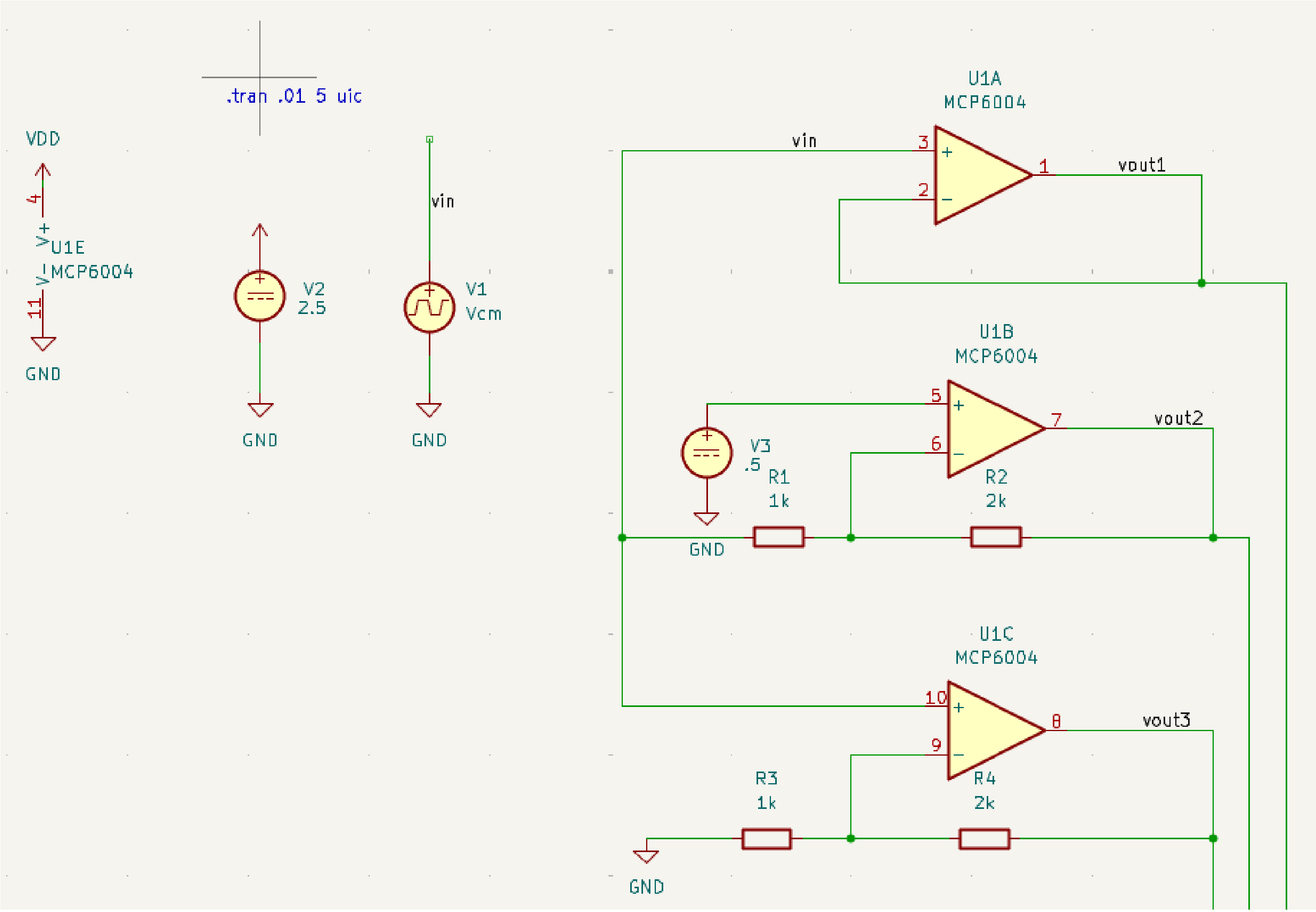
Q4.2: Does it amplifiy V_{cm}?

Lesson 2: complex circuits – amplifiers – feedback

KK5: try the following configurations for the amplifier.

Q5.1:
derive the three functions $V_{out}(V_{in})$.
Treat the Op like a voltage controlled $(V_+ - V_-)$ ideal voltage source.

Hint:
Assume, that the amplification A of the OP is so high (~10^6) that the difference between the two inputs is nearly zero whenever the output voltage is not gnd or vdd. Use this fact to answer the question, i.e. use
$(V_+ - V_-)=0$
and calculate $V_{out}(V_{in})$ using Kirchhoffs laws.
The input resistance of the + and – pins is so high that you can assume $I_+=0$ and $I_- = 0$.



Lesson 2: complex circuits – amplifiers – adder circuit

KK6: the three output voltages are combined by resistors at the noninverting input of the U1D.

Q6.1: derive the function of the circuit and verify in simulation